

What is an intelligent system?

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Abstract

The concept of intelligent system has emerged in information technology as a type of system derived from successful applications of artificial intelligence. The goal of this paper is to give a general description of an intelligent system, which integrates previous approaches and takes into account recent advances in artificial intelligence. The paper describes an intelligent system in a generic way, identifying its main properties and functional components, and presents some common categories. The presented description follows a practical approach to be used by system engineers. Its generality and its use is illustrated with real-world system examples and related with artificial intelligence methods.

1 Introduction

Mankind has made significant progress through the development of increasingly powerful and sophisticated tools. In the age of the industrial revolution, a large number of tools were built as machines that automated tasks requiring physical effort. In the digital age, computer-based tools are being created to automate tasks that require mental effort. The capabilities of these tools have been progressively increased to perform tasks that require more and more intelligence. This evolution has generated a type of tool that we call intelligent system.

Intelligent systems help us performing specialized tasks in professional domains such as medical diagnosis (e.g., recognize tumors on x-ray images) or airport management (e.g., generate a new assignment of airport gates in the presence of an incident). They can also perform for us tedious tasks (e.g., autonomous car driving or house cleaning) or dangerous tasks such as exploration of unknown areas (e.g., underwater exploration).

An intelligent system is therefore a tool designed to perform tasks that require intelligence. The development of such a type of system is now an engineering discipline of information technology that requires effective methods and tools. The precise characterization of an intelligent system is non trivial because it is based on terms related to cognition, an area that is not fully understood and admits different interpretations. Some of the used terms can even change with the proposal of new computational models of intelligence and new scientific findings related to our understanding of the mind.

The main purpose of this paper is to present a characterization of an intelligent system that integrates and updates previous conceptions of such type of system. It follows a pragmatic approach to be useful in an engineering context to help system engineers conceive, analyze and build intelligent systems. It is defined as a design metaphor identifying the main general functional components and properties.

The remainder of the paper presents the following contents. Section 2 presents a definition of intelligent system with the description of its properties and its main functional components. Section 3 describes classes of intelligent systems using the presented characterization, covering recent achievements of this technology. Section 4 describes aspects in the development process of an intelligent system such a types of artificial intelligence methods used to develop the different components of intelligent systems,

Definition 1: Intelligent agents are computational systems that inhabit some complex dynamic environment, sense and act autonomously in this environment, and by doing so realize a set of goals or tasks for which they are designed (Maes 1995)
Definition 2: Intelligent agents continuously perform three functions: perception of dynamic conditions in the environment; action to affect conditions in the environment; and reasoning to interpret perceptions, solve problems, draw inferences, and determine actions (Hayes-Roth 1995)
Definition 3: Intelligent agents operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals (Russell and Norvig 2014)
Definition 4: Intelligent agents are autonomous or semi-autonomous hardware or software systems that perform tasks in complex, dynamically changing environments. Agents communicate with their environment and effect changes in their environment by executing actions (Muller 1996)

Table 1: Sample of definitions of intelligent agent.

2 Definition of intelligent system

Intelligent systems have been characterized in the literature of AI using the concept of *intelligent agent* (Wooldridge and Jennings 1995) (Franklin and Graesser 1996) (Russell and Norvig 2014). This concept provides an adequate degree of abstraction that helps to identify general properties. Table 1 shows a sample of definitions of intelligent system considered as an agent. These definitions highlight that the system works in an environment, has a set of capabilities (perception, learning, etc.), and makes decisions about how to act.

The word *system* (instead of agent) is used by some authors (e.g., Meystel and Albus (Meystel and Albus 2002)) and in academic areas of information technology such as the names of university courses or academic journals (e.g., IEEE Intelligent Systems and International Journal of Intelligent and Robotic Systems). In this case, the word *system* emphasizes the presence of multiple components that must be adequately combined to create intelligence. Knowing how to do this combination efficiently is one of the key aspects in the development of such type of systems.

The characterization of an intelligent system used in this paper integrates parts of the previous agent-based definitions. For example, the identification of a separate complex dynamic world is important to give an adequate operational context, at a certain degree of abstraction. This paper also uses a set of cognitive abilities (e.g., perceive, reason, learn, etc.) usually enumerated by agent-based definitions.

However, this paper characterizes the mentioned capabilities separated in two distinguished parts. On the one hand, we distinguish a primary set of cognitive abilities to interact with the world including perception, action control, deliberation or language use. On the other hand, the system has a secondary set of abilities about how to use the primary abilities, resulting in a more complex intelligent behavior. These second abilities are related to (1) rationality to maximize system performance, (2) improving behavior through learning, and (3) observation through introspection which allows, for example, explaining the use of the own knowledge.

According to this, the definition that we follow in this paper as a basic scheme to structure the further characterization of an intelligent system is presented below. The following sections describe in more detail each one of the three parts of the definition. It is important to note that, this characterization is not based on rigid definitions that establish whether a system is intelligent or not. Instead, it should be considered flexible to be adapted when it is used by system engineers according to their particular needs in the analysis or development of specific applications.

Definition. An intelligent system is a tool that (1) operates in a complex world with limited resources (2) possesses primary cognitive abilities such as perception, action control, reasoning, or language use, and (3) exhibits complex intelligent behavior supported by abilities such as rationality, adaptation through learning, or the ability to explain the use of its knowledge by introspection.

2.1 Property 1: Working in a complex world

As mentioned in the previous sections, one of the properties of an intelligent system is that it is a tool designed to perform tasks in a complex world using limited resources. For example, this means that the complexity of the world is significantly higher than the information processing capacity of the system. The amount of components of the environment, their interactions or the anticipated effect of potential actions in the environment cannot usually be completely represented in the

memory of the intelligent system. For example, in the chess game, it is not possible to anticipate the exact effect of all potential movements, because the amount of combinations is too high.

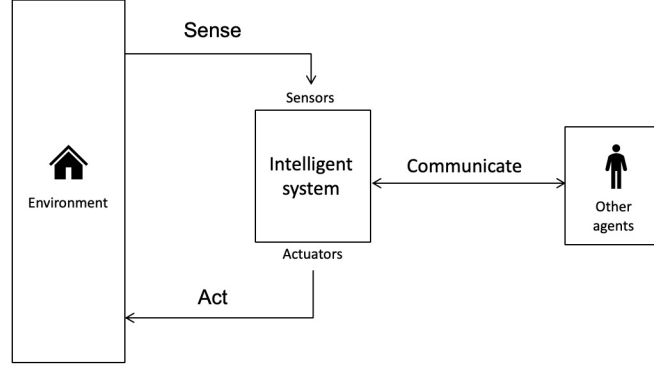


Figure 1: World interaction of an intelligent system.

We can distinguish two types of components in the world: the *environment* and *other agents* (human agents or artificial computer-based agents) (Figure 1). The system observes features from the environment through sensors and performs actions using actuators.

Example. A thermostat is a simple example that illustrates the concept of a system that operates in an environment (e.g. house rooms). The thermostat uses a thermometer to sense the temperature of the room, and actuates by starting or stopping a heater. The thermostat also communicates with a human user who starts and stops the thermostat, or establishes the desired temperature.

The use of sensors and actuators (real or virtual) separates clearly the body of the intelligent system from the rest of the environment. This characteristic is called *embodiment*. It is also said that the agent is *situated* in an environment when the agent operates in a close-coupled continuous interaction with environment in a continuous sequence of sense-act.

The degree of complexity of the environment with respect to the agent can be described with the following features (Wooldridge 2009; Russell and Norvig 2014):

- *Static* (or *dynamic*), the environment does not change (or changes) while an agent is deliberating,
- *Discrete* (or *continuous*), the state of the environment, time, percepts or actions are discrete (or continuous),
- *Fully-observable* (or *partial-observable*), sensors detect (or do not detect) all aspects that are relevant to the choice of action,
- *Deterministic* (or *stochastic*), the next state of the environment is (or it is not) completely determined by the current state and the action,
- *Episodic* (or *sequential*), actions do not influence (or they influence) future actions,
- *Known* (or *unknown*), the outcomes for all actions are known (or they are not known) by the agent in advance.

Example. The environment for a chess player is static, discrete, fully observable, deterministic, sequential and known. In the case of a self-driving car, the environment is continuous, partial observable, stochastic, sequential and known.

2.2 Property 2: Primary cognitive abilities

We distinguish four primary cognitive abilities of an intelligent system (Figure 2): (1) *perception*, ability to extract relevant information from the environment¹, (2) *deliberation*, ability to reason

¹Note that the general term environment may include parts of the own body (arms, legs, etc.) that can also be sensed by the system. This type of perception is called *proprioception* (sense of self-movement and body position).

about the world and make decisions about what to do, (3) *interaction*, ability to communicate with other agents using language, and (4) *action control*, i.e., ability to control the execution of the own actions. Perception and actuation may be divided in multiple separate components that operate in parallel at a high frequency. However, deliberation and communication should operate sequentially which is necessary for coherent reasoning (Laird, Lebiere, and Rosenbloom 2017) and consistent communication.

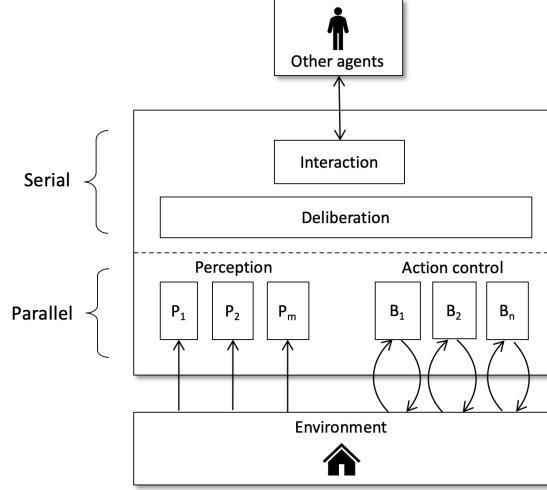


Figure 2: Main functional components of an intelligent system.

Example. A self-driving car is an example of intelligent system that illustrates the four primary cognitive abilities. The car senses the world with the help of multiple sensors (lidar, cameras, accelerometers, acoustic sensors, etc.). *Perception* is achieved with multiple perceptual components that extract specific information about the world. Each perceptual component is specialized in the recognition of a kind of world object (e.g., pedestrians, traffic signals, other vehicles, etc.). *Action control* is supported by several behaviors that control the motion of the vehicle such as steering, braking, accelerating, etc. Each behavior is controlled separately by an actuation controller. Action control components operate in a closed continuous loop with the environment requiring also data from specialized sensors. For example, the component that controls acceleration operates observing the vehicle speed. Concerning *deliberation*, the autonomous car is able to generate a path to reach a destination by planning. Finally, *interaction* is established with the passenger in such a way that the passenger may indicate the destination to go and the system presents the path to follow and the information about the environment that helps the passenger monitor the correct execution of the trip.

2.2.1 Deliberative and reactive

Deliberative behavior contrasts with reactive behavior. In general, deliberation (Ingrand and Ghallab 2017) includes multiple functions such as planning or goal reasoning. This paper assumes that the fundamental characteristic of deliberation is the ability to reason about the world and reach justifiable conclusions. On the other hand, *reactive* behavior generates instantaneous actions in response to a stimulus. This reactive behavior is present, for example, in animal reflexes or in behaviors based on human intuitions, which reach conclusions based on feelings rather than on facts or evidence.

Reactive behavior is useful because it allows an intelligent agent to react efficiently to dynamic events of the environment. A situated agent may work with reactive behaviors with a limited or nonexistent memory about the world. The immediate feedback of the state of the world through sensors can be used to make quick decisions and, consequently, react quickly to events.

Deliberative behavior is related to *rational thinking*. An agent that thinks rationally makes decisions that are consistent with its own knowledge (with its own beliefs and goals) and, therefore, the agent is able to find reasons (i.e., a logical justification) for the decisions taken. In contrast to reactive agents, deliberative agents with rational thinking can explain the reasons that support a

decision. Deliberation may be also useful to inhibit reactive behaviors in some cases, when it is necessary to postpone immediate answers in order to reach more useful long-term goals.

The separation in this two ways of thinking is related to the human mind as it is described, for example, by Daniel Kahneman (Kahneman 2011) with the names System 1 and System 2. System 1 is slow, logical, effortful, conscious, linguistic (can be described verbally) and makes complex decisions. On the contrary, system 2 is fast, intuitive, automatic, unconscious, non-linguistic (difficult to verbalize) and corresponds to everyday decisions.

In the development of computer-based intelligent systems, bridging the gap between these two parts is in general a non trivial problem that require specialized functions (e.g., attention mechanisms, symbol grounding, cognizant failure and action meta-control).

2.2.2 Interaction with other agents

The type of interaction with other agents is an important characteristic that distinguish different types of intelligent systems. The interaction may be established with only one single agent (e.g., an end user that delegates tasks or asks for advice) or with multiple agents organizing several intelligent systems in complex structures in the form of multi-agent systems that may cooperate using social coordination mechanisms to achieve common goals.

During the interaction, an intelligent system may be *proactive*². This means that the system do not need to wait until a user asks to do a task, but it takes the initiative to perform tasks based on what the system perceives from the world and its own goals.

Autonomy is a relevant characteristic that an intelligent system may exhibit during the interaction with other agents. In general, an autonomous system makes its own decisions about how to act in the environment to complete a task delegated by another agent. As figure 3 illustrates, autonomous systems can be contrasted with systems that are designed to give advice. An autonomous system *acts in the world to help the user*. This means that the user delegates in the system a task and the system decides and performs autonomously specific actions in the environment that allow the task to be completed.

In contrast to this, an intelligent advisor system *helps the user act in the world*. In this case, the user is the one who makes the decisions about what actions to do and the role of the system is to give advice by providing useful information to facilitate decisions. This information can be generated using descriptive or predictive analytical methods (e.g., pattern recognition, methods for diagnosis or temporal projection). In addition, the system can use prescriptive methods (e.g., resource assignment, scheduling or planning) to recommend what actions should be done .

An autonomous system can also accept or reject proposed tasks from the user according to certain reasons, such as the current situation of the environment or its own goals (e.g., safety goals). For this purpose, the system can verify the correctness and feasibility of a requested task before it is performed and explain to the user the reasons that justify why a requested task is rejected.

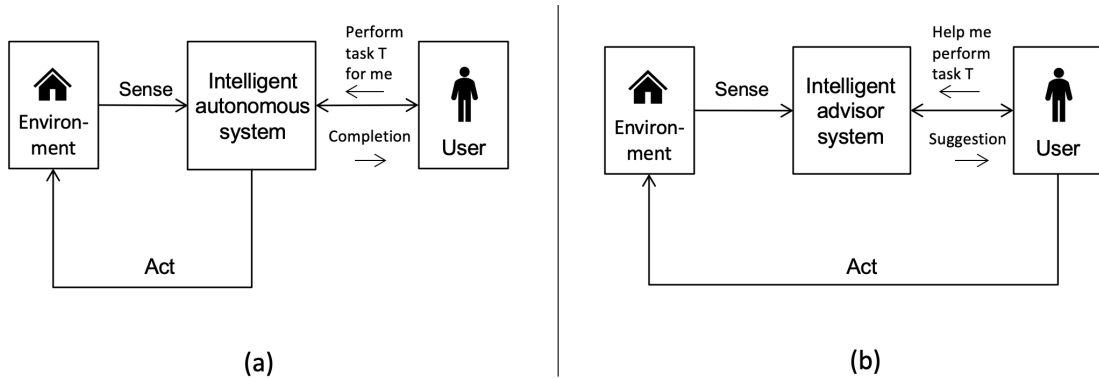


Figure 3: Comparison between (a) autonomous system and (b) advisor system.

²Some authors use the term *reactive* as the opposite of *proactive*. However, in this paper we will use the term *reactive* as the opposite of *deliberative* as it is generally used in agent-based systems.

2.3 Property 3: Complex intelligent behavior

A system can exhibit complex intelligent behavior with certain abilities such as acting rationally, adaptation through learning, or introspection. The following sections describe these abilities in more detail.

2.3.1 Acting rationally

An intelligent system that acts rationally³ uses its cognitive abilities in the best possible way to obtain the maximum performance. In the chess game, for example, a performance measure can be the number of tournaments won by the chess player. In a medical diagnostic system, a performance measure can be the number of times that diseases are correctly diagnosed. The performance measure of the recommendations of an investment advisor may be the economic profit obtained.

Since the environment is usually complex, an intelligent system is not able to determine in advance the exact performance measure that will be obtained, but estimations based on expected performance with uncertainty. This can be explicitly programmed using algorithms that make decisions based on the expected performance using for example decision theory. In other cases, rationality may be implicit in the algorithms because it is imposed by the design decisions made by the developers when they build the intelligent system.

Rational decisions may affect different cognitive abilities. For example, a rational agent may decide what parts of the observed environment require more attention (perception) or what is the next question to ask to a user during information gathering (interaction). The rational agent may also decide what is the most appropriate method for solving a task or what is the best next action to do (deliberation) or the best way to control actions (actuation).

2.3.2 Adaptation through learning

An agent with the capacity of *learning* is able to improve its performance in the course of multiple interactions with the world. For example, a chess player can learn more effective strategies after playing multiple games. This capacity allows the intelligent system to *adapt* to the world and its changes.

Learning can improve different cognitive abilities. For example, an aerial robot can learn to land more precisely on a marked surface after multiple trials (action control). A personal assistant can improve its speech recognition after listening to multiple sentences from the same voice of a user (perception).

An intelligent system that is able to learn needs a value judgement method that evaluates its behavior. The result of such evaluation is used by the intelligent system to change its internal model to modify its future behavior in order to obtain more positive evaluation. The value judgement method can be an *external* agent who works as a instructor (or trainer or supervisor) observing and assessing the behavior of the intelligent system to provide positive and negative feedback (e.g., in the form of reward or punishment). The intelligent system can also use an *internal* method that judges its own actions based, for example, on simulated feelings (e.g., pain or pleasure etc.).

2.3.3 Introspection

An intelligent system with the capacity of introspection can analyze how operate its own cognitive abilities. This means that the system has a self-reflective ability supported by an observable model that represents dynamically how its own cognitive abilities work. Such a model can be used as input data of algorithms to simulate self-awareness processes.

One of the important practical utilities of introspection is that it allows the system to generate explanations to an external agent (e.g., a human end-user) to give credibility in the system's conclusions. For example, in the case of an intelligent advisor system that recommends decisions to an end-user, the system can generate convincing logical justifications connected with evidences that allow end-users to assume the responsibility of the decisions.

³Note that *acting rationally* and *thinking rationally* are considered in this paper as two different functionalities.

3 A taxonomy of intelligent systems

The definition used in this paper for an intelligent system enumerates a set of cognitive abilities such as perception, action control, interaction, or deliberation. The presence or absence of such abilities may be used to identify different types of systems. Figure 4 organizes in a hierarchical structure some of the main types of intelligent systems. This taxonomy is not intended to be accurate and exhaustive but to introduce existing systems highlighting their main differences. The presented structure should be considered approximate because it includes certain simplifications to present the categories in a tree structure.

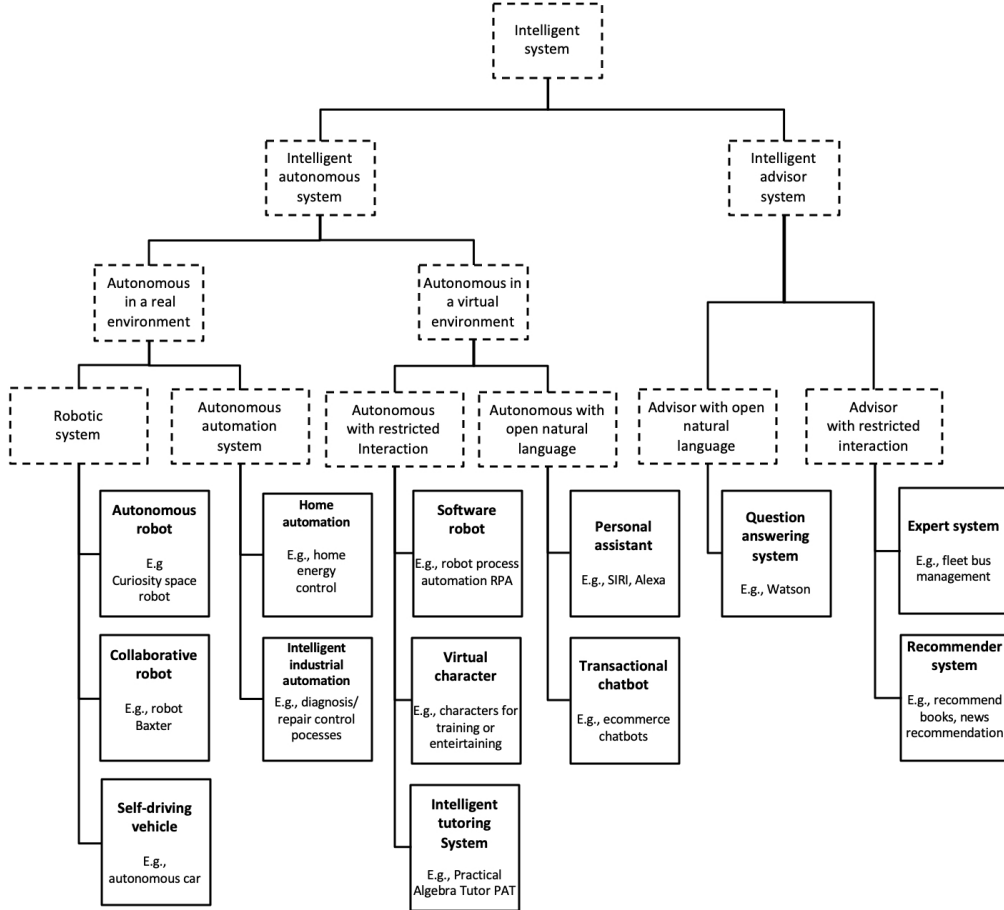


Figure 4: A taxonomy of intelligent systems.

The following sections describe in more detail some types of these intelligent systems and illustrate them with real-world examples showing how they can be described with the characterization presented in this paper.

3.1 Expert systems

The development of expert systems contributed to the practical success of AI in the 1980s with many commercial systems in multiple domains. They were applied to perform automatically tasks such as the following:

- *Classification*: Find the category of an object using observations (e.g., determine the presence of a mineral, recommend a type of investment, detect abnormal behaviors).
- *Diagnosis*: Find the causes of symptoms (e.g., medical diagnosis, mechanical diagnosis).
- *Temporal projection*: Estimate the future behavior of a dynamic system (e.g., economic market prediction).

- *Assignment/Scheduling*: Assign a set of resources to a set of needs (e.g., assign airplanes to gates in a airport).
- *Configuration*: Create a design that satisfy some requirements (e.g., design the machinery of an elevator).
- *Planning*: Find a set of actions to reach a desired state (e.g., emergency evacuation planner)

3.1.1 First generation expert systems

The first generation of expert systems used mainly heuristic reasoning with symbolic knowledge representations such as rules. For example, Mycin was one of the first expert systems that was used successfully as a model in other types of problems. Mycin was developed at the Stanford university in the 1970s (Buchanan and Shortliffe 1984) and it was able to diagnose infectious diseases and propose therapies. This system has the following two main functional components:

- *Interaction*. The information about patients is obtained through the communication with the user. Dialogue with the user is done in a question-answering process with prefixed questions and answers. Mycin asks questions about the patient and cultures. The questions are adapted to the previous answers (questions have context). The user may answer with uncertainty and may ask for explanations. Mycin presents diagnosis and therapy recommendation
- *Deliberation*. Mycin diagnoses the causes of the symptoms finding the possible infectious organisms. Mycin uses a heuristic classification method with a knowledge base with 450 rules and backward chaining. Mycin includes an original method for approximate reasoning (certainty factors as values in $[-1, +1]$). In addition, Mycin recommends therapies using a method based on a generate and test procedure.

The methods used for building expert systems demonstrated that they are valid in many problems, but they present limitations to be used in certain cases. One of the most important limitation is what is called the *knowledge acquisition bottleneck*. Expert systems use heuristic knowledge formalized with declarative representations (e.g., rules and frames) that is stored in a knowledge base. The creation and maintenance of a knowledge base usually requires significant effort. Therefore, this approach is limited for the development of knowledge bases that are not too large.

Expert systems may be able to solve difficult problems, but they operate in very specialized domains, without general knowledge about the rest of the world. Therefore, compared to human experts, who have also common sense knowledge, expert systems are brittle and unable to react correctly in unexpected situations.

Another problem is related to the way these systems obtain information about the environment. For example, Mycin gets information about the patient using a prefixed set of questions and answers. This communication mechanism may be too narrow and rigid to be used in situations when the format of available data is more diverse (e.g., unstructured texts). This limitation makes difficult to integrate the expert system into day-to-day operations.

3.1.2 Expert systems for the management of sensorized environments

The behavior of dynamic environments can be monitored with the help of sensor networks and information systems. This is the case, for example, of modern information infrastructures that have been developed in the last decades applied for example to strategic areas such a smart cities (transportation, climate, pollution surveillance) or industry 4.0 (with sensorized industrial installations).

In this context, it has emerged the need of using intelligent systems that advise operators to make decisions for the management of such environments. There are multiple examples of expert systems that belong to this category. For instance, there are systems for public transportation management (Molina 2005), scheduling and coordination at an airport (Jo et al. 2000), emergency decision support in floods (Molina and Blasco 2003) or electric power operation (Filho et al. 2012).

The first example corresponds to an expert system that detects incidents and recommends actuation in a urban bus transportation network. This system was included in 2002 as part of the fleet management system of the bus control center of the city of Vitoria in Spain. This system has the following main abilities:

- *Perception.* The system collects data from buses using GPS localization and localizes the position of each bus in a map of transportation lines.
- *Deliberation.* The system detects the presence of incidents (e.g., detection of delays based on current position and planned position) and predicts future behavior (e.g., a model is used about historic demand for transportation). The system determines actions to be done to manage incidents (an automated planner based on hierarchical task network (HTN) planning is used to generate action plans).
- *Interaction.* The system presents to the operator detected incidents (e.g., bus delays). The operator can notify other events to the system (e. g., a broken bus, a blocked street, etc.). The system recommends actions (e.g., using additional buses, sending repairing truck, etc.). The system justifies actions (e.g., why some lines are prioritized based on expected demand).

In general, this type of systems include a *perception* component to observe and interpret characteristics of the environment behavior using a sensor network or an information system. In addition, the system includes a *deliberation* component to derive conclusions from data, such as:

- what happens in the environment (detection),
- why it happens (diagnosis),
- what may happen in the future (temporal projection, what-if analysis), and
- what should be done (planning).

The system also includes a *interaction* component to present this information to the human operator. This type of intelligent system does not have an *actuation* component to modify the environment. The operator is responsible of making management decisions and implementing the actions in the environment. Therefore, it is important that the interaction component presents the system suggestions with sufficient justification in such a way that the operator can take responsibility for the decisions.

3.2 Question-answering systems based on cognitive computing

The company IBM developed a system called Watson (Ferrucci et al. 2010) that marked a significant milestone in the history of artificial intelligence when it won the first prize of the quiz TV show "Jeopardy" in 2011 competing against human opponents. The system was able to respond better than humans to open domain questions written in natural language about general culture.

In the quiz game, Watson had to guess an answer based on a given clue provided by the user. For example, Watson receives a clue such as "this drug has been shown to relieve the symptoms of ADD with relatively few side effects". In less than 5 seconds, Watson had to generate the answer expressed as: "what is Ritalin?" (according to the rules of the game, the answer "Ritalin" is presented as a question).

Watson used a selected large corpus of text documents of about 2 million pages as a source of information (e.g., news, encyclopedias and literary works). Watson searches candidate answers in such a corpus that can be possible responses to questions formulated by the end user in natural language. Watson uses a sophisticated structure of algorithms to (1) select relevant search keywords that may be used to find out (partial) candidate partial answers and (2) rank and integrate partial answers to generate the final answer in natural language.

The general characteristics of Watson can be described using to the components presented in this paper. In our characterization, we may consider the corpus of documents as the environment in which the system operates to perform its searching task. According to this, the components of Watson are the following:

- *Interaction.* The interaction component of Watson uses natural language processing to extract the linguistic structure of the given clue. The interaction component also generates the final answer in natural language (in form of a question).
- *Deliberation.* Deliberation in Watson included two parts. A first part generates a number of search keys from the linguistic structure of the given clue. Each search key is generated to be used later (by the perception component) to retrieve information from text documents.

The second part of deliberation ranks and combines retrieved candidate answers. For this purpose, Watson is able to merge answers (Kalyanpur et al. 2012) doing temporal reasoning, geospatial reasoning and taxonomic reasoning (subsumption, disjointness, etc.) using ontologies and databases (e.g., Yago, DBPedia, Freebase). In order to score answers, Watson uses models generated by machine learning.

- *Perception.* To obtain candidate answers from the corpus of documents, Watson used dozens of specialized algorithms for information extraction using natural language processing.

Watson is a successful representative case of a question-answering system that uses a computing approach called *cognitive computing*. The idea of this approach is that the system uses as a source of information large amounts of structured and non-structured data (text, images, etc.). The system performs automatically cognitive tasks that include feature extraction (e.g., using natural language processing or computer vision), answer aggregation (reasoning with ontologies or knowledge bases), natural language communication (using natural language understanding and generation) and evidence assignment of hypotheses using models learned by machine learning.

3.3 Autonomous robots

An autonomous robot may be also considered as a representative case of intelligent system. An example of this system is an aerial robot for airplane inspection developed as a demonstrative prototype in 2019 for the company Airbus (Bavle 2019). This system performs the following functions:

- *Perception.* The aerial robot is equipped with multiple sensors such as (1) inertial measurement unit (IMU) to provide information about orientation, angular velocities, and linear accelerations, (2) cameras to calculate velocity using visual odometry, (3) a lidar sensor that generates a 3D point cloud, and (4) a high resolution frontal camera. The perception system performs data fusion from multiple sensors for a precise localization. The high resolution camera is used to detect anomalies on the surface of the airplane using computer vision.
- *Deliberation.* In this system, the main deliberative function is path planning. The system may generate an inspection plan to cover the surface of the airplane. In addition, the system stores images to be analyzed later.
- *Action control.* Action control corresponds to plan execution and motion control to complete the navigation plan. Action control also includes a reactive mechanism to avoid unexpected obstacles detected with the help of the lidar sensor.
- *Interaction.* The operator provides the geometry of the airplane and initiates the mission. The operator can stop the mission execution if an emergency happens.

In general, an intelligent robotic system is one of the most complex category of intelligent system because it usually includes the majority of the components considered in this paper and, in addition, it operates in real-world environments with physical constraints. In this category, there are multiple kinds of systems with different kinds of complexity such as industrial collaborative robots (e.g., Baxter), autonomous vehicles (e.g., self driving cars, planet explorer Curiosity), assistive robots, domestic robots (e.g., house cleaning robot Roomba), entertaining robots (e.g., Sony Aibo, Furby), etc.

4 Building intelligent systems

This section summarizes two of the important aspects in the construction of an intelligent system: (1) the selection of artificial intelligence methods and (2) the acquisition of the domain knowledge used by the system to perform the desired tasks.

General cognitive ability	Example function	Examples of detailed functions	Examples of computational methods
Perception	Feature extraction	Image recognition, information extraction, signal processing, natural language understanding, attention mechanisms	Convolutional neural networks, natural language processing (rules, statistic models, LSTM neural networks), statistical data analysis
	Data aggregation	Data abstraction, data fusion.	Statistical data analysis
Deliberation	Semantic memory	Knowledge retrieval, knowledge integration, belief revision, symbol grounding	Ontologies, truth maintenance systems, perceptual anchoring
	Reasoning	Analytic reasoning (diagnosis, classification, etc.), synthetic reasoning (planning, assignment, etc.), goal reasoning	Probabilistic reasoning, automated deduction, taxonomic reasoning, automated planning, constraint satisfaction, heuristic search
Interaction	Language use	Natural language understanding, natural language generation, speech recognition	Natural language processing (rules, statistic models, LSTM neural networks)
	Coordination	Dialog management, social coordination.	Multi-agent coordination algorithms
Action control	Action selection	Goal-driven action selection, event-driven action selection, action conflict management	Behavior trees, Petri nets, finite-state machines
	Action execution control	Motion control, manipulation control	Reinforcement learning, neural networks, fuzzy control, control theory

Table 2: Functionalities related to cognitive abilities together with examples of computational methods used for building intelligent systems.

4.1 Artificial intelligence methods

The development of a complex intelligent system can require using several artificial intelligence methods. Table 2 shows functionalities related to cognitive abilities together with examples of some of common methods used for building intelligent systems.

These methods can be divided according to two distinguished approaches. One approach uses symbolic methods such as logic-based representations, taxonomic representations or constraints based representations. The other approach uses non-symbolic methods such as connectionist representations (e.g., artificial neural networks).

Symbolic and connectionist methods correspond to alternative approaches that have complementary strengths and weaknesses (Franklin 1995). In fact, there is a lack of understanding of how information processing performed by symbolic approaches can be mapped onto computations performed by connectionist methods. The term *computational explanatory gap* (Reggia 2014) has been used to express that it is unclear how the two approaches are related.

Cognitive abilities that are consciously accessible (such as processes related to deliberation or interaction) may be implemented using symbolic methods. These methods can describe the symbols that represent knowledge to justify how to reach a certain conclusion.

On the other hand, perception and action control are usually carried out in an unconscious way and they are better implemented with non-symbolic approaches such as connectionist methods. In practice, it is interesting to note that the implementation of unconscious sensorimotor skills require much more computation compared to deliberative reasoning. This has been stated as the *Moravec’s paradox* (Moravec 1988).

One of the difficulties in the development of complex intelligent systems is the combination of deliberation (with symbolic approaches) and reaction (with connectionist approaches). This may require additional mechanisms such as perceptual anchoring for symbol grounding (Coradeschi and Saffiotti 2002) or action execution control with cognizant failure.

4.2 Knowledge acquisition

Intelligent systems normally use *knowledge models* that represent knowledge that is difficult to be formalized using procedural representations of computer languages such as Java, Python or C++.

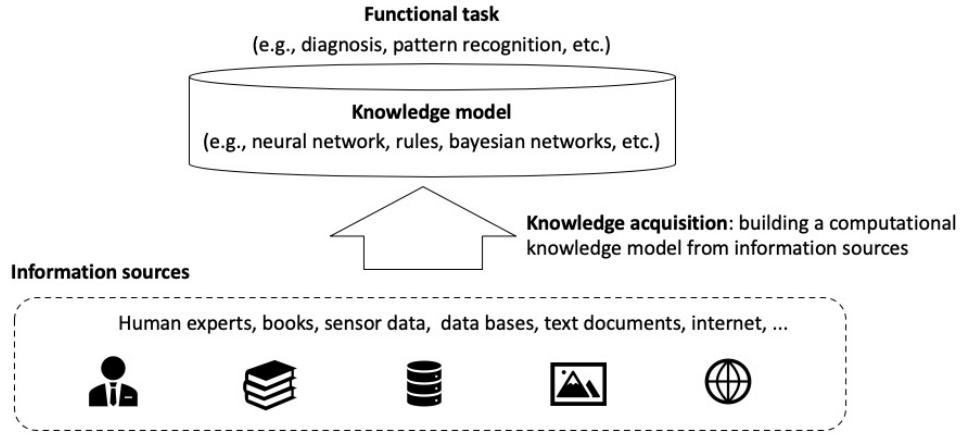


Figure 5: Overview of the knowledge acquisition process.

These models are used by general inference algorithms to generate conclusions from given premises.

Example. Consider an example of a medical expert system that uses a model that relates symptoms and diseases. This model, which is called *knowledge base*, is represented using symbolic rules. An inference algorithm processes such rules to diagnose the disease of a given patient. The inference algorithm is general and, therefore, it may be implemented using generic software tools. However, the rules of the knowledge model are specific and must be written for the particular expert system.

The construction of a knowledge model by developers is called *knowledge acquisition* (figure 5) and can be done either manually or automatically. Many of the knowledge bases of classical expert systems (like the Mycin medical system) represent heuristic knowledge that was acquired manually from domain experts and formalized using declarative representations (e.g., rules or first order logic).

The creation and maintenance of such knowledge bases usually requires significant effort. This has been called *knowledge acquisition bottleneck* because it is one of the most difficult tasks of the development of the intelligent system. To facilitate this work, ontological engineering can be applied to reuse part of the content of knowledge bases across different intelligent systems.

In contrast to manual acquisition, it is possible to apply automatic methods for knowledge acquisition which has gained popularity in the last decades due to the increasing availability of large amounts of data. Automatic acquisition methods can use information sources with different formats such as images (or videos), structured data (e.g., data bases with alphanumeric data, temporal series) or non-structured text (e.g., written in natural language).

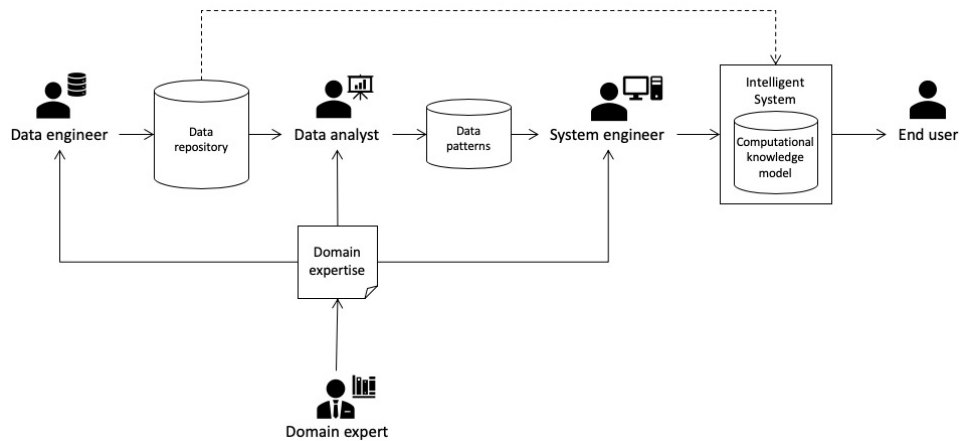


Figure 6: Summary of the participants in the development of an intelligent system. The dashed arrow at the top of the figure means that the data repository could be used as input of the intelligent system.

In order to acquire knowledge automatically, it is possible to use machine learning or natural language processing. Machine learning methods can be used, for example, to generate knowledge models from examples. For instance, a neural network can be trained to recognize defects in images of a surface of an airplane. Natural language processing methods can be used to extract knowledge from text documents written in natural language.

The construction of intelligent systems may involve multiple participants that can be summarized in the following main roles:

- *System engineer.* The goal of the system engineer is building the final computational implementation of the intelligent system. The main specialization area of system engineers is computer science (knowledge representation, machine learning, natural language processing, software engineering, computer programming) and they may be familiar with tools and computer language such as TensorFlow, Python, Prolog, Drools, ROS and C++.
- *Data analyst.* The goal of the data analyst is discovering and describing patterns in data (statistical analysis, predictive analysis, etc.) that will be part of the intelligent system. Data analysts are mainly specialized in Mathematics (statistics, data visualization, machine learning algorithms) and related software tools such as R, Python, SAS, Matlab and Weka.
- *Data engineer.* The goal of the data engineer is building efficient and accessible data repositories to be used for data analysis or as information source of the intelligent system. Data engineers are specialized in data base management (conventional data bases, distributed data bases, big data architectures) and software tools and languages such as SQL, Hadoop, MongoDB.
- *Domain expert.* In the development of an intelligent system helps, the presence of domain experts are important to help the rest of participants providing the required support about domain knowledge (e.g., biology, medicine, finance, economy, etc.).

5 Conclusions

This paper has presented a characterization of an intelligent system to be used in an engineering context that may be used by system engineers to conceive, analyze and build this type of systems. The presented definition describes main functional components that are illustrated in the paper by analyzing the characteristics of different categories of intelligent systems.

The description it is based on terms related to common understandings of cognition in artificial intelligence. As a future work, we expect to adapt and extend this general characterization to keep its consistency with new results derived from further research about new computational models of intelligence.

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